

Ziheng Ji

Target Roles: End-to-End Algorithms for Autonomous Driving / Robotics
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EDUCATION

Shanghai Jiao Tong University, M.Eng. in Electronic Information, Shanghai Sep. 2024 - Mar. 2027
Global Institute of Technology; Advisor: Tong Qin; Research: End-to-End Policy Learning / Sim-to-Real
Beijing Institute of Technology, B.Eng. in Vehicle Engineering, Beijing Sep. 2020 - Jul. 2024
School of Mechanical Engineering; admitted to SJTU through graduate recommendation

TECHNICAL SKILLS

Algorithms and Models: PyTorch, Transformers, BEV / LSS, trajectory generation, imitation learning, reinforcement learning, Flow Matching
Simulation and Deployment: 3DGS, Isaac Sim, MuJoCo, ROS, Docker, Ubuntu, closed-loop vehicle and robot testing
Programming and Languages: Python, Git, MATLAB / Simulink; CET-6 528, TOEFL iBT 100

PROJECT EXPERIENCE

Real-Scene Digital Twins for Policy Training and Closed-Loop Evaluation (Scene2Gym) 2025 - 2026

- Led 3DGS scene reconstruction, object-scale alignment, and joint physics-rendering simulation. Built a Sim-to-Real pipeline covering real-scene capture, synthetic data generation, policy training, and closed-loop real-robot evaluation.
- In unseen real scenes, imitation learning reached **70% to 82%** real-robot success and visual RL reached **56% zero-shot success**. A native simulator-rendering baseline reached **0%**. Accepted to **IROS 2026**; first author.

End-to-End Vision-Based Parking System (ParkingE2E) Jan. 2024 - Oct. 2024

- Designed an end-to-end parking system that takes surround-view images and a target parking space as input. Led visual data collection, model training, trajectory output, and closed-loop vehicle deployment across the camera-to-planning-to-control stack.
- Built multi-view BEV features with LSS, fused them with a target-slot encoding, and designed a Transformer-based autoregressive trajectory generator for multistep prediction and closed-loop control.
- Tested the system in underground garages and outdoor parking lots, achieving an approximately **85% parking success rate**. Published at **IROS 2024 (Oral)**; second author.

Scene Reconstruction and Robot Simulation 2025 - Present

- Worked on scene reconstruction and interactive simulation environments for robot learning, including scene assembly and physical consistency checks.
- Further details of ongoing work are withheld during anonymous review.

VLA Policy Reproduction and Bimanual Robot Deployment (π^* 0.6 Recap) 2025 - Present

- Built a real-robot framework for data collection, policy training, and inference. Reproduced the two-stage pipeline of imitation learning pretraining and reinforcement learning post-training, then deployed it for long-horizon bimanual manipulation of deformable objects.
- The deployed policy reached an approximately **70% real-robot success rate** from randomized, cluttered garment configurations, covering the full workflow from data collection and training to inference and RL fine-tuning.

INTERNSHIP EXPERIENCE

Zhuoyu (Incoming) Expected start: Sep. 21, 2026
Xiaomi, Embodied AI Division - Algorithm Engineer Apr. 2026 - Aug. 6, 2026

- Built a policy training and evaluation platform using Isaac Sim and photorealistic 3DGS rendering. Developed pipelines for real-scene reconstruction, automated scene generation, and data synthesis to support large-scale policy training and closed-loop evaluation.
- Produced simulated scenes and task data, contributed to imitation learning pretraining and policy deployment, and completed closed-loop real-robot validation on the AgileX Cobot Magic dual-arm robot, connecting synthetic data generation with on-robot evaluation.

CATL, Intelligent Manufacturing Division - Algorithm Engineer Sep. 2025 - Dec. 2025

- Designed a single-arm manipulation pipeline for coolant-line connector insertion. Combined CAD priors, visual pose estimation, semantic segmentation, and visuotactile sensing to develop and test the insertion policy on hardware.

PUBLICATIONS

Scene2Gym: Real-World Scenes to Interactive Robot Training Environments First author, IROS 2026
ParkingE2E: Camera-based End-to-end Parking Network, from Images to Planning Second author, IROS 2024 (Oral)
4 manuscripts under review; details withheld during anonymous review.

COMPETITION EXPERIENCE

Beijing Institute of Technology Formula Student Driverless Team 2020 - 2024

- National Champion, Formula Student Driverless China (2021, 2022).** Contributed to the autonomous racecar from drive-by-wire chassis development through perception and localization.
- Served as Team Captain in 2024, coordinating vehicle development and testing while contributing to SLAM and perception. Served as Mechanical Team Lead from 2021 to 2023, responsible for steer-by-wire design and vehicle-dynamics simulation.

HONORS AND AWARDS

Graduate Scholarship, Second Class (2025); Undergraduate Scholarship, Second Class (2021, 2022); National Level-I Athlete, Badminton (2020-2025)